



Adaptive Data-Driven Audio Network for Speech Deepfake Detection

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Outline



- Introduction
- Methods
- Results
- Summary

Audio deepfake detection

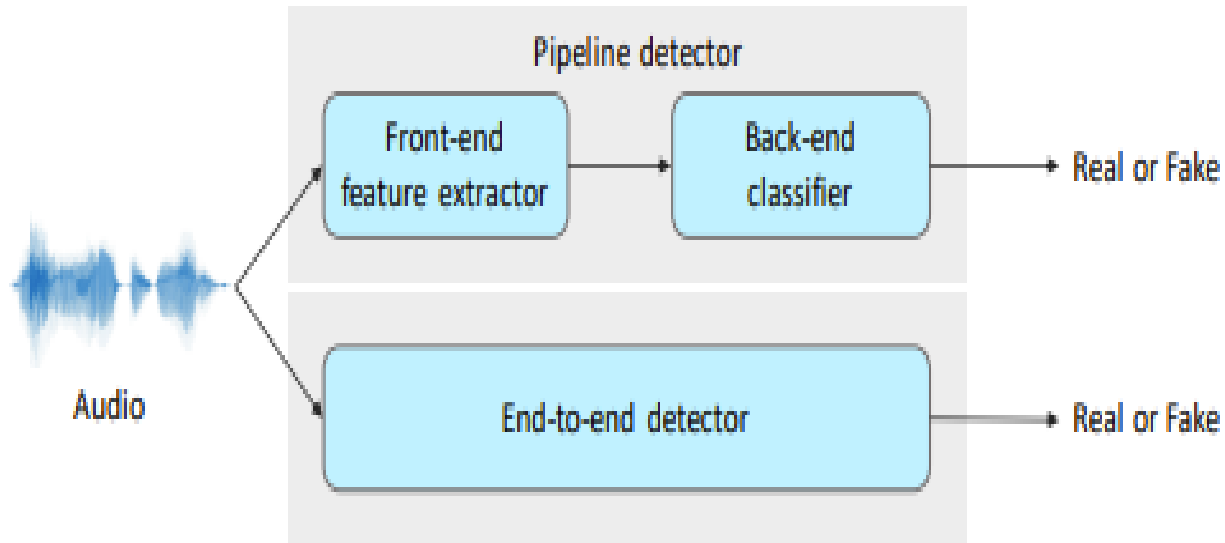


Figure: Audio deepfake detection structure [1]

Goal:

- Combat misinformation and fraud
- Protect the automated speaker verification system

Potential applications:

- Cybersecurity and fraud prevention
- Digital forensics
- Secure authentication (Biometrics)

Current Audio Deepfake Detection Models

AASIST/ AASIST 3 (and variants)

- 1D raw audio data as input
- SincConv and residual blocks in the models.
- Similar to RawNet2 structure.
- Graph-based attention network
- Combined with SSL frontends

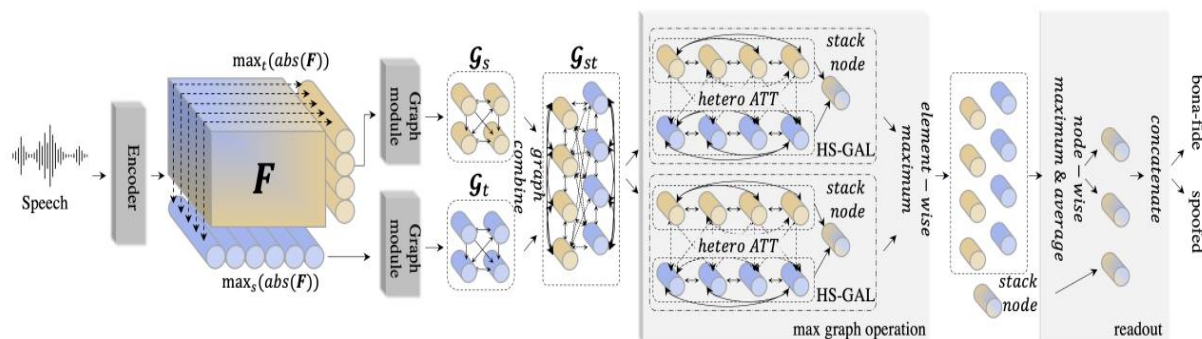


Figure: AASIST structure [2]

Wav2Vec2, XLS-R, or HuBERT

- Raw waveform as input
- Multiple layers of CNN to generate features
- Latent representations are fed into a transformer network

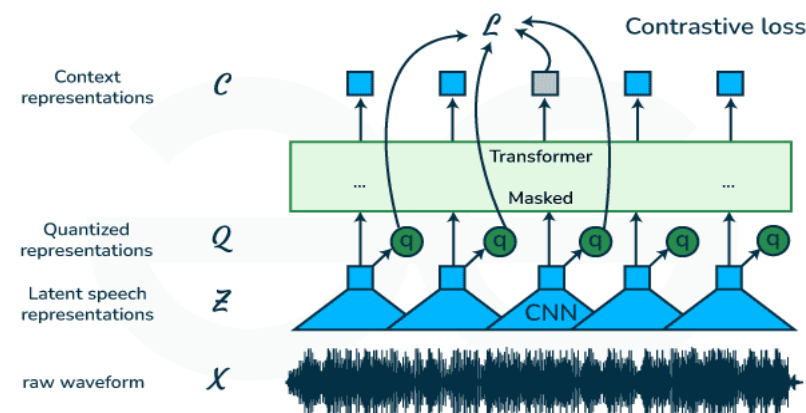


Figure: Wav2Vec2 structure [3]

SincConv Layer in AASIST

- SincConv: 1D convolutions with band-pass filters
- Risk of overfitting to frequency bands
- Parameter reduced for convolution filter coefficients, but higher on kernel generation (5-10%) FLOPs contribution

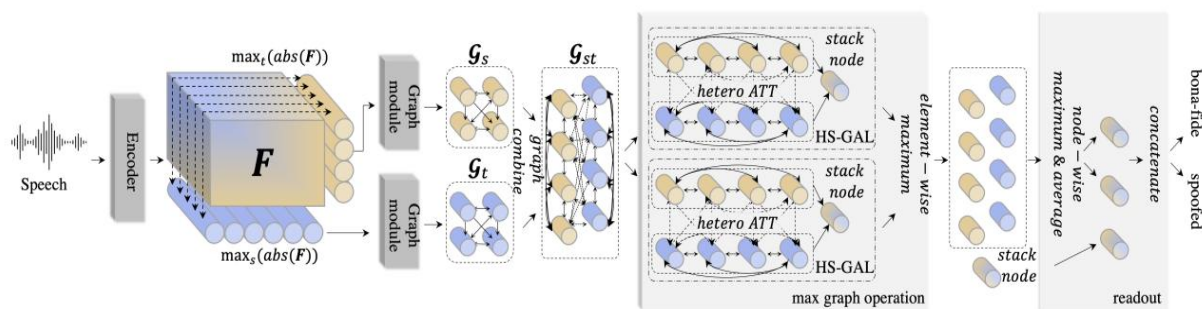
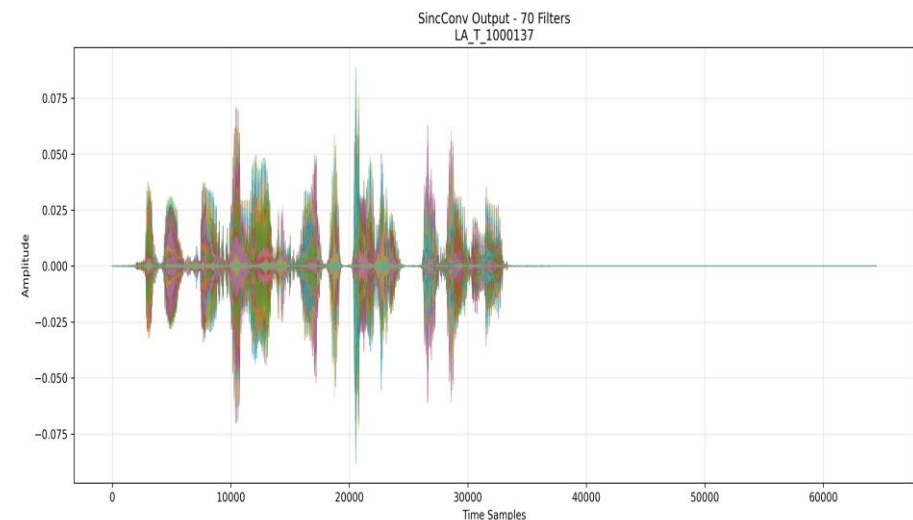
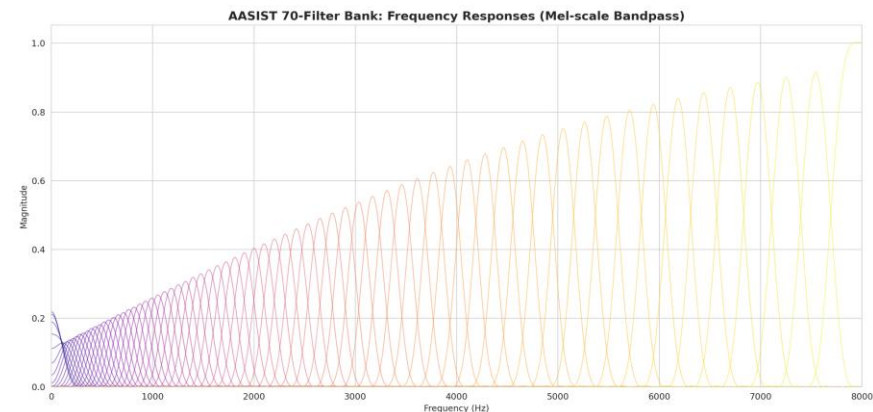


Figure: AASIST structure [2]



Residual Blocks in AASIST

- SincConv: 1D convolutions with band-pass filters
- Risk of overfitting to frequency bands
- Parameter reduced for convolution filter coefficients, but higher on kernel generation (5-10%) FLOPs contribution
- Residual block as encoder is computationally heavy
 - About 70% FLOPs contribution
- Far lighter than WavLM, XLS-R in memory

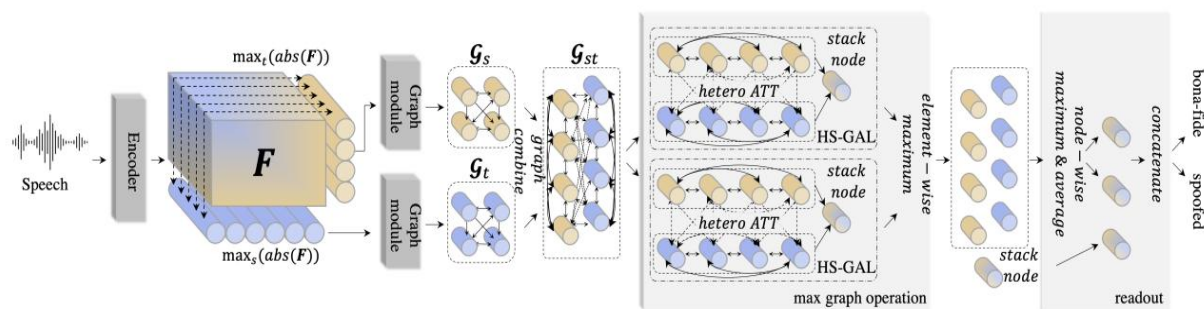
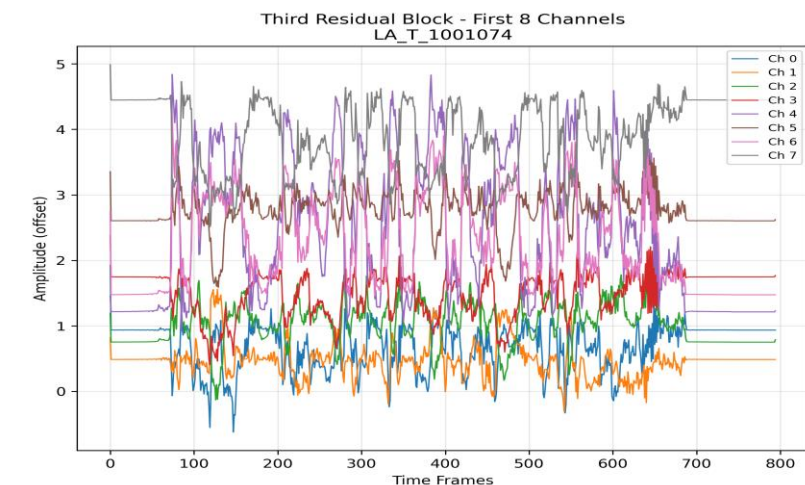
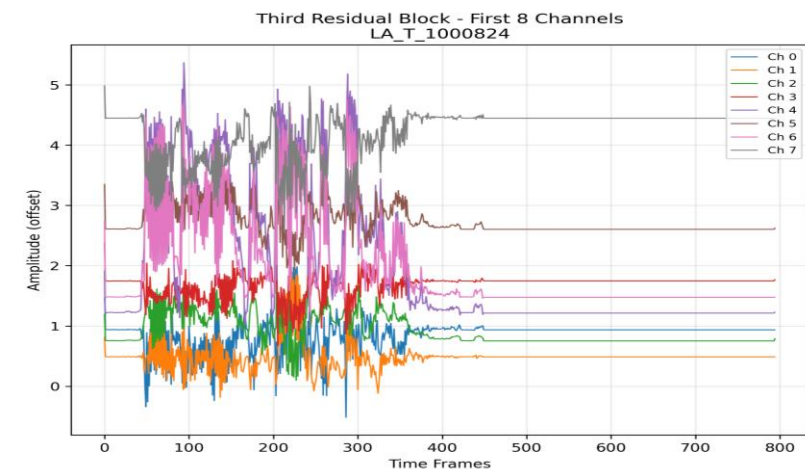
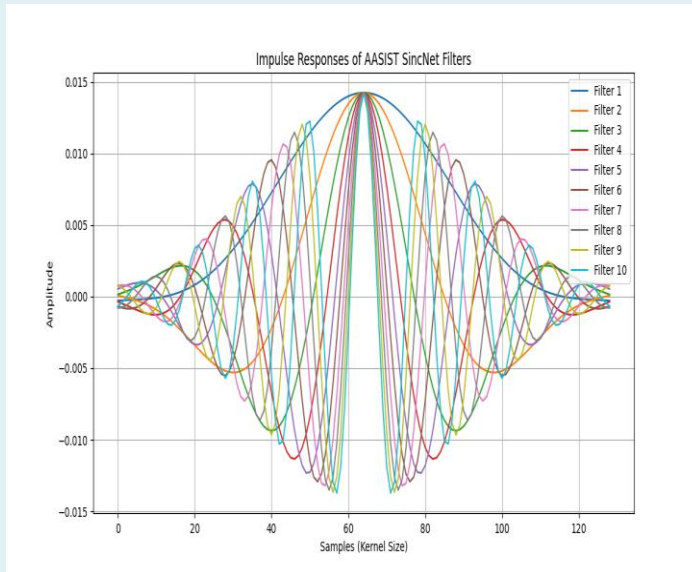


Figure: AASIST structure [2]

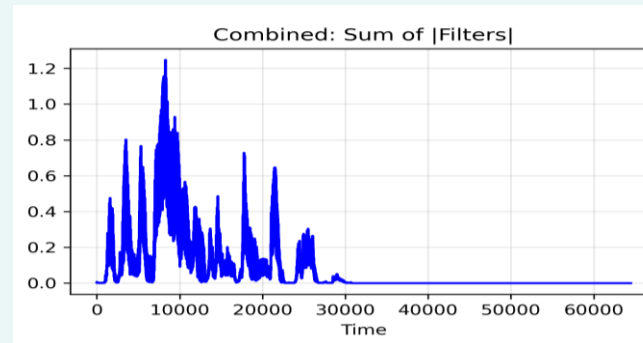
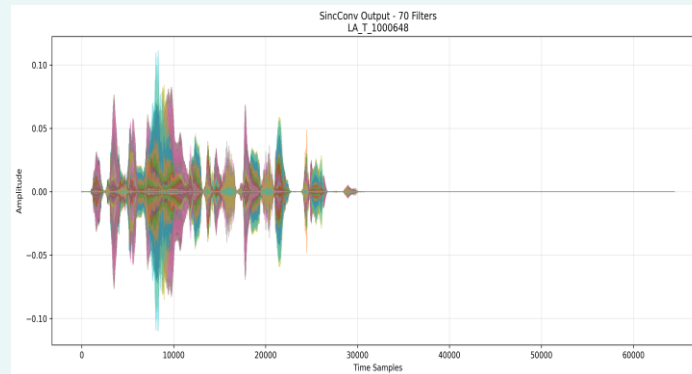


SincConv filters

Impulse response of SincConv filters



LA_1000648 output waveform



SincConv filters

- Applied in the time domain to preserve the phase information.
- Stride=1, preserves maximum temporal resolution (compared with stride=4-8 in wav2vec2 CNN)
- Detect synthetic artifacts at fine time scales
- Downsample in residual blocks

Experimental Results

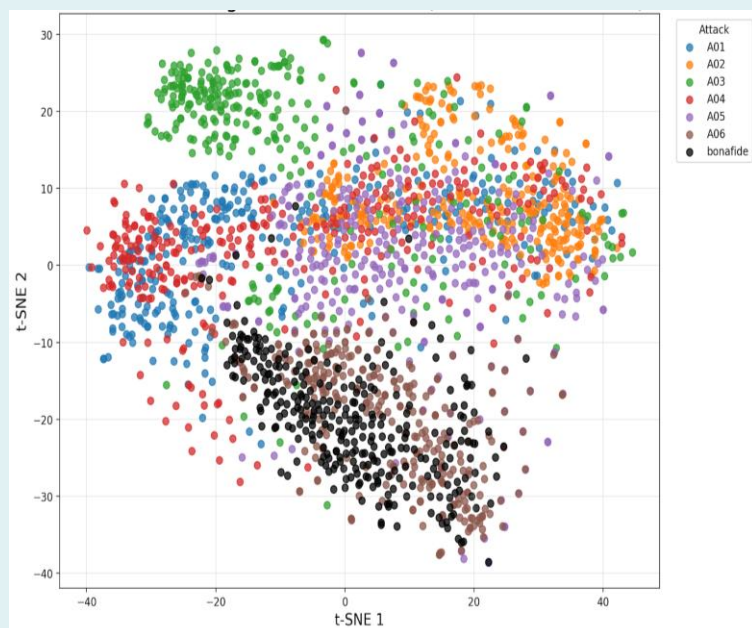
- Datasets
 - English: ASVspoof 2019 LA dataset [4]
 - 107 speakers. 10000 true speech data, and 10000 fake data using 19 methods.
 - Mandarin: CFAD dataset (Chinese dataset for fake audio detection). R01-R04 real audio data and the clean fake dataset for testing [5]
 - Use true speech data from 3 source, 6 types of fake data with different AI methods.
 - Cantonese: the fake data are generated with methods: AR, Minimax, Cantonese AI, Vits2, HifiGAN, and Bvs2; the real data obtained from Tan Lee's team at CUHK.
 - 10,000 data samples, true data 5000 samples, and AI generated data 5000 samples using 6 methods.
 - Fake audio data examples: AR, BVS2, CantoneseAI, HifiGAN, Minimax, and VITS2; true speech data example.

Experimental Results

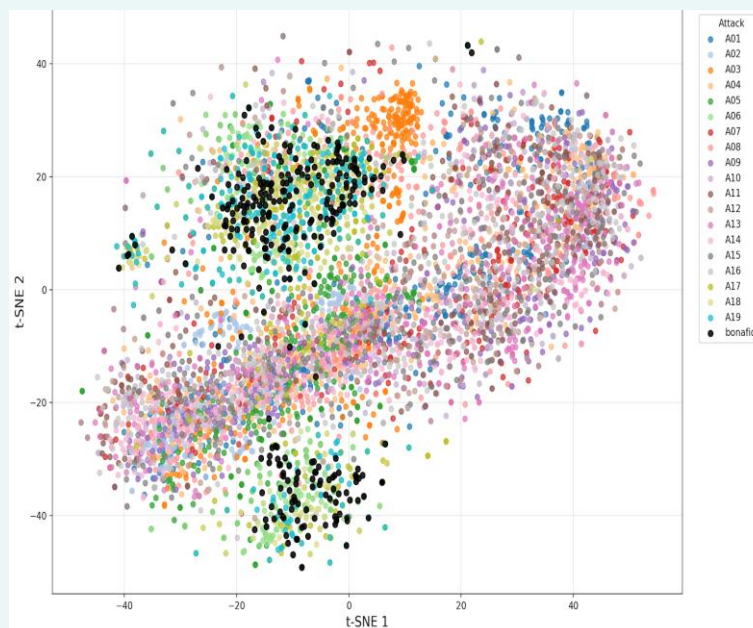
- Evaluation:
 - Classification accuracy
 - Clustering results

ASVspoof 2019 Experiment

Clustering with T-SNE after PCA projection into 100 dimensions of real and fake data (6 classes)



Clustering with T-SNE after PCA projection into 100 dimensions of real and fake data (19 classes)

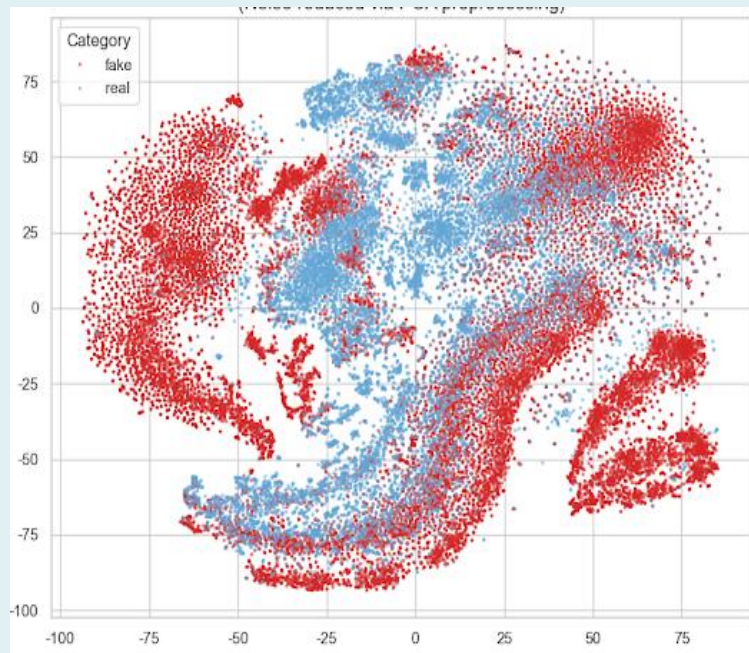


Results

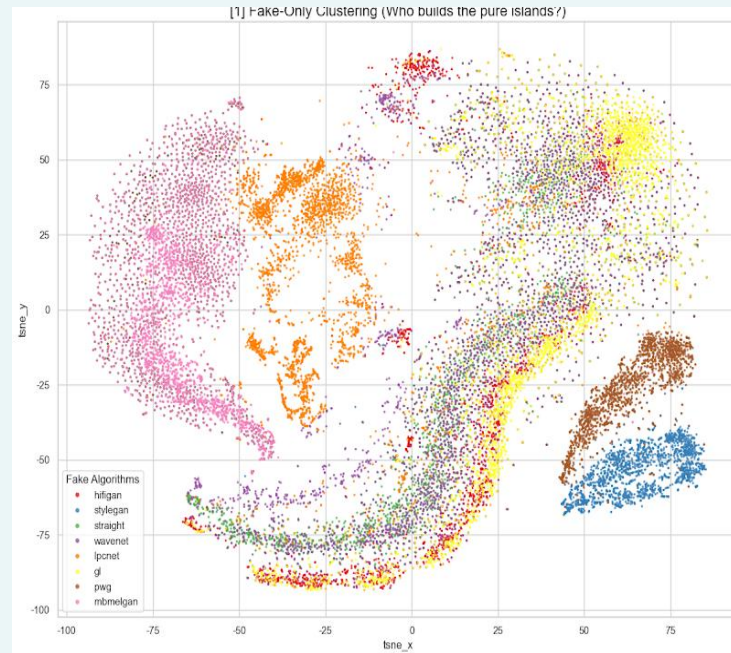
- Sampling frequency: 16k, truncated into 3.5 seconds for preprocessing.
- Training 80% of all data, validate 20%. 5-fold cross validation.
- Classification accuracy calculation, with 90% to distinguish real and fake audio; and 85% to distinguish different fake audio generation methods.
- Baseline: CQCC+GMM accuracy 76.17%

CFAD Experiment (Mandarin)

Clustering with T-SNE after PCA projection into 50 dimensions of real and fake data



Clustering with T-SNE after PCA projection of fake data using different methods

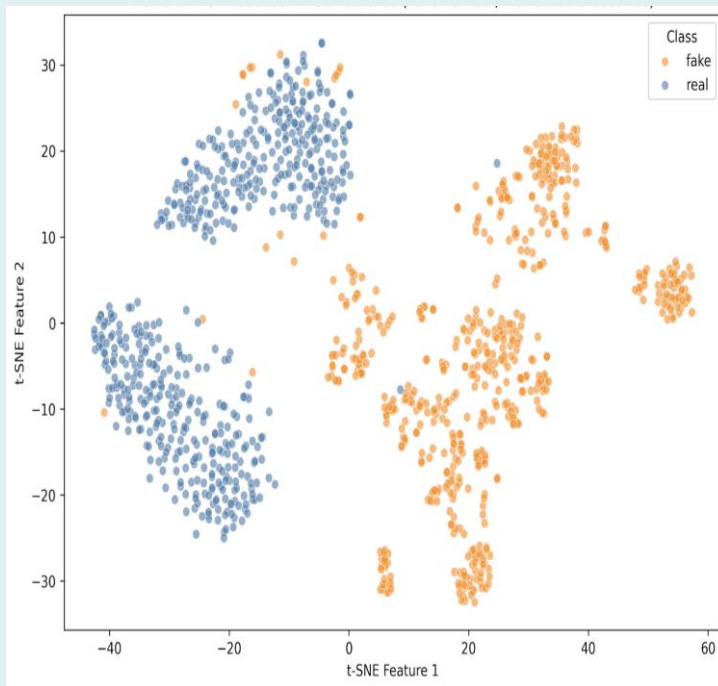


Results

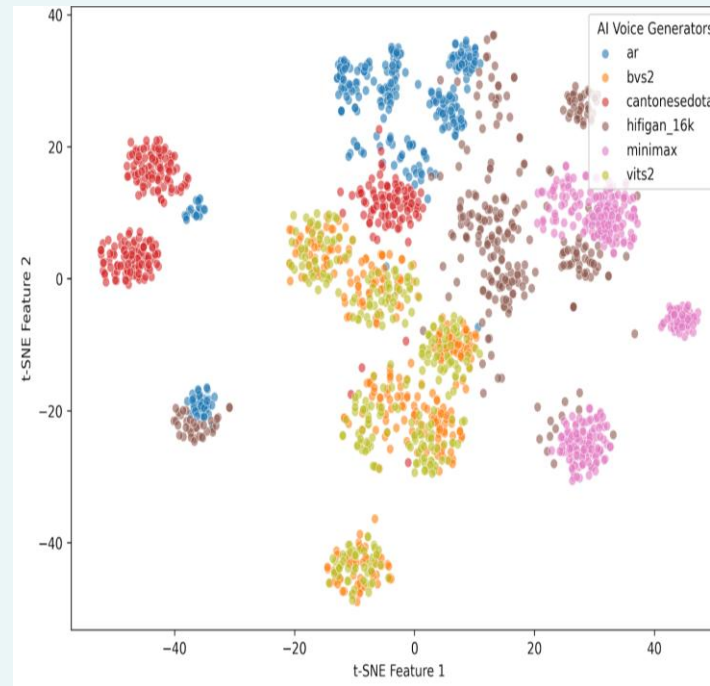
- Sampling frequency 22k, truncated into 3.5 seconds in length in preprocessing.
- Training 80% of all data, validate 20%. 5-fold cross validation.
- Classification accuracy calculation, with **95%** to distinguish real and fake audio; and 92% to distinguish different fake audio generation methods.
- Baseline: LFCC+LCNN accuracy 76.1%.

Experiment with Cantonese speech

Clustering with T-SNE after PCA projection into 100 dimensions of real and fake data



Clustering with T-SNE after PCA projection of fake data using different methods



Results:

- Cantonese Speech dataset
 - Truncate them in 3 seconds for preprocessing, sampling frequency 22k.
 - 6 AI methods to generate fake data, each has 300 samples.
- Training 80% of all data, validate 20%. 5-fold cross validation.
- Classification accuracy calculation, with 99% to distinguish real and fake audio; and 83% to distinguish different fake audio generation methods.

Major Reference

- [1] Yi, J., Wang, C., Tao, J., Zhang, X., Zhang, C. Y., & Zhao, Y. (2023). Audio deepfake detection: A survey. *arXiv preprint arXiv:2308.14970*.
- [2] Jung, J. W., Heo, H. S., Tak, H., Shim, H. J., Chung, J. S., Lee, B. J., ... & Evans, N. (2022, May). AASIST: Audio anti-spoofing using integrated spectro-temporal graph attention networks. In *ICASSP 2022-2022 IEEE international conference on acoustics, speech and signal processing (ICASSP)* (pp. 6367-6371). IEEE.
- [3] Liu, A. H., Hsu, W. N., Auli, M., & Baevski, A. (2023, January). Towards end-to-end unsupervised speech recognition. In *2022 IEEE Spoken Language Technology Workshop (SLT)* (pp. 221-228). IEEE.
- [4] Nautsch, A., Wang, X., Evans, N., Kinnunen, T. H., Vestman, V., Todisco, M., ... & Lee, K. A. (2021). ASVspoof 2019: Spoofing countermeasures for the detection of synthesized, converted and replayed speech. *IEEE Transactions on Biometrics, Behavior, and Identity Science*, 3(2), 252-265.
- [5] Ma, H., Yi, J., Wang, C., Yan, X., Tao, J., Wang, T., ... & Fu, R. (2024). CFAD: A Chinese dataset for fake audio detection. *Speech Communication*, 164, 103122.
- [6] Xian, Y., Pu, Y., Gan, Z., Lu, L., & Thompson, A. (2017, March). Adaptive DCTNet for audio signal classification. In *2017 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)* (pp. 3999-4003). IEEE.
- [7] Cai, J. F., Ji, H., Shen, Z., & Ye, G. B. (2014). Data-driven tight frame construction and image denoising. *Applied and Computational Harmonic Analysis*, 37(1), 89-105.
- [8] Xian, Y., Gu, H., Wang, W., Huang, X., Yao, Y., Wang, Y., & Cai, J. F. (2018, November). Data-driven tight frame for cryo-em image denoising and conformational classification. In *2018 IEEE Global Conference on Signal and Information Processing (GlobalSIP)* (pp. 544-548). IEEE.



THANK YOU

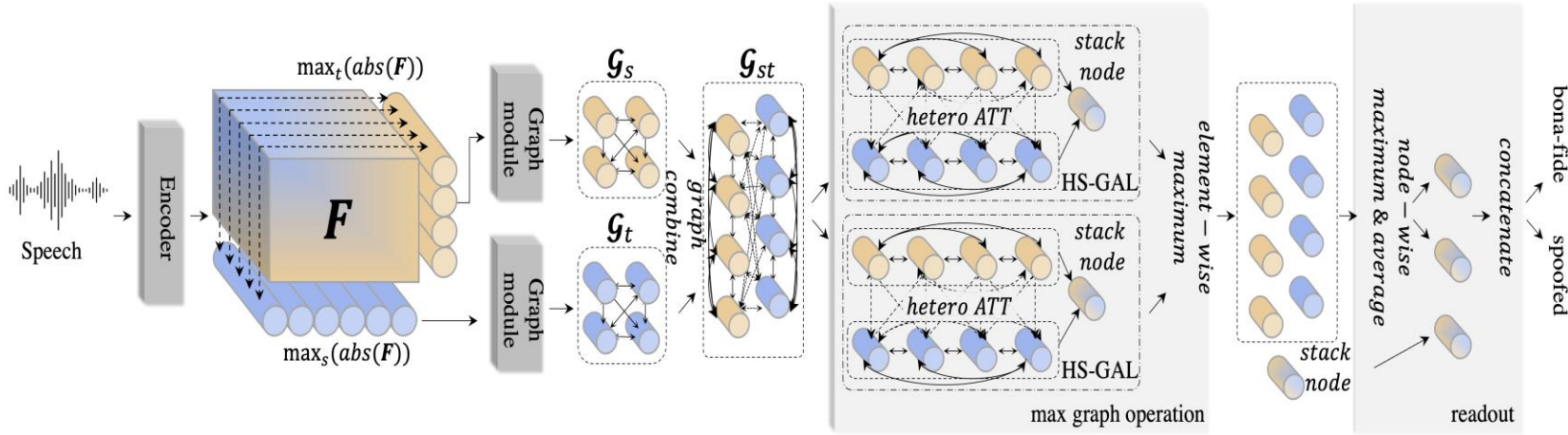
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Overall pipeline of AASIST

$$\begin{aligned}x &\xrightarrow{\text{SincConv}} \mathbf{Y} \in \mathbb{R}^{B \times 70 \times T'} \\ \mathbf{Y} &\xrightarrow{\text{unsqueeze}} \mathbf{X}_0 \in \mathbb{R}^{B \times 1 \times 70 \times T'} \\ \mathbf{X}_0 &\xrightarrow{\text{MaxPool} + \text{BN} + \text{SELU}} \mathbf{X}_3 \\ \mathbf{X}_3 &\xrightarrow{6 \text{ Residual Blocks}} \mathbf{F} \in \mathbb{R}^{B \times 64 \times S' \times T''} \\ \mathbf{F} &\xrightarrow{\text{Max over dims} + \text{pos}} \mathbf{X}_S, \mathbf{X}_T \\ \mathbf{X}_S, \mathbf{X}_T &\xrightarrow{\text{HS-GAT} + \text{MGO}} \text{graph embeddings} \\ &\xrightarrow{\text{Readout} + \text{FC}} \text{logits}\end{aligned}$$

Overall pipeline of AASIST



$$\begin{aligned}
 x &\xrightarrow{\text{SincConv}} \mathbf{Y} \in \mathbb{R}^{B \times 70 \times T'} \\
 \mathbf{Y} &\xrightarrow{\text{unsqueeze}} \mathbf{X}_0 \in \mathbb{R}^{B \times 1 \times 70 \times T'} \\
 \mathbf{X}_0 &\xrightarrow{\text{MaxPool} + \text{BN} + \text{SELU}} \mathbf{X}_3 \\
 \mathbf{X}_3 &\xrightarrow{6 \text{ Residual Blocks}} \mathbf{F} \in \mathbb{R}^{B \times 64 \times S' \times T''} \\
 \mathbf{F} &\xrightarrow{\text{Max over dims} + \text{pos}} \mathbf{X}_S, \mathbf{X}_T \\
 \mathbf{X}_S, \mathbf{X}_T &\xrightarrow{\text{HS-GAT} + \text{MGO}} \text{graph embeddings} \\
 &\xrightarrow{\text{Readout} + \text{FC}} \text{logits}
 \end{aligned}$$